Predicate Logic Conti….

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Properties of Statements:

- **Satisfiable:** A statement is satisfiable if there is some interpretation for **which it is true**.
- **Contradiction:** A sentence is contradictory (unsatisfiable) if there is **no interpretation** for **which, it is true**.
- **Valid:** A sentence is valid **if it is true for every interpretation**.
- **Equivalence:** Two sentences are equivalent if they have the **same truth value** under every interpretations.

At least one TRUE Satisfiable

Always FALSE Contradiction

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Conti...

A statement Q is a logical consequence of P if:
Whenever P is TRUE, Q must also be TRUE.

- **Logical Consequences:** A sentence is a logical consequence of another if it is satisfied by all interpretations which satisfy the first.
- **A valid statement is satisfiable, and a contradictory statement is invalid.**

Example: On The above definitions:

- **P** is satisfiable but **not valid** since an interpretation **that assigns false to P assigns false to the sentence P**.
- **P V ~P** is **valid** since every interpretation results in a value of **true for (P V ~P).**
- **P & ~P** is a **contradiction** since every interpretation results in a value of **false for (P & ~P)**.

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Conti…

- **P and ~(~P)** are **equivalent** since each has the **same truth values** under every interpretation.
- **P** is a **logical consequence** of (P & Q) since any interpretation for which **(P & Q) is true**, **P is also true**.

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Theorem

AND of all s is true

- **Theorem 4.1: The sentence s is a logical consequence of s_1 , s_2 , …… , s_n if and only if s_1 & s_2 & s_3 . . . . & s_n → s is valid.**
- **Proof:** Theorem 4.1 can be seen by first noting that if s is a logical consequence of s_1 , s_2 , …… , s_n , then for any interpretation ***I*** in which s_1 & s_2 & s_3 , ….. . &s_n → s is true.
- on the other hand, if s_1 & s_2 & s_3 , …. & s_n →s is valid, then for any interpretation ***I*** if s_1 & s_2 & s_3 , ….. . &s_n is true, **s is also true**.

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Theorem

- **Theorem 4.2:** The sentence s is a logical consequence of s_1 , s_2 , ... , s_n if and only if **s_1 & s_2 & s_3 , ... & s_n & ~s** is **inconsistent**.

- **Proof:** The proof of theorem 4.2 follows directly from **theorem 4.1** since s is a logical consequence of s_1 , s_2 , ... , s_n if and only if s_1 & s_2 & s_3 , ... &s_n → s is valid, that is, if and only if **~**(s_1 & s_2 & s_3 , ... &s_n → s) is inconsistent.

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Conti .....

- But

$\sim(s_1 \,\&\, s_2 \,\&\, s_3 , \ldots . \,\&s_n \rightarrow s) = \sim(\sim(s_1 \,\&\, s_2 \,\&\, s_3 , \ldots . \,\&s_n) \vee s)$

[**By Conditional Elimination**]

$= \sim\sim(s_1 \,\&\, s_2 \,\&\, s_3 , \ldots . \,\&s_n) \,\&\, \sim s)$

[**By De Morgan's Law**]

$= s_1 \,\&\, s_2 \,\&\, s_3 , \ldots . \,\&s_n \,\&\, \sim s$

- When s is a logical consequence of $s_1 , s_2 , \ldots , s_n$, the formula $s_1 \,\&\, s_2 \,\&\, s_3 , \ldots . \,\&s_n \rightarrow s$ is called a theorem, with s is the conclusion.
- When **s** is a logical consequence of the set **S** = $\{s_1 , s_2 , \ldots , s_n\}$ we will also set **S** logically implies or logically entails **S** , written $S \vdash s$.

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Table 4.2 lists some of the important laws of PL (Some Equivalence Laws)

Name of Laws	Statements
Idempotency	P V P = P P & P = P
Associativity	(PVQ)VR = P V (Q V R) (P & Q) & R = P & (Q & R)
Commutativity	P V Q = Q V P P & Q = Q & P P ↔Q = Q↔P
Distributivity	P & (Q V R) = (P & Q)V (P &R) P V (Q & R) = (P V Q) & (P V R)
De Morgan's Laws	~(P V Q) = ~P & ~Q ~(P & Q) = ~P V ~Q
Conditional Elimination	P → Q = ~P V Q
Bi-conditional Elimination	P ↔Q = (p → Q) & (Q →P)

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Example

- **Show that P→Q is equivalent to ~PVQ and that P↔Q is equivalent to the expression (P→Q)&(Q→P)**.

- The truth table 4.3 is given bellow.

TABLE 4.3 : Truth table for equitant sentences

P	Q	~P	(~PVQ)	(P→Q)	(Q→P)	(P→Q)& (Q→P)
T	T	F	T	T	T	T
T	F	F	F	F	T	F
F	T	T	T	T	F	F
F	F	T	T	T	T	T

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Inference Rules

- **The inference rules of PL provide the means to perform logical proofs or deductions.**

Inference rules tell us what we can logically conclude from what we already know.

- Few Such Rules are as follows:
  - ❑**Modus ponens**
  - ❑**Chain Rule**

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Modus Ponens:

- From P and P → Q infer Q. This sometimes written as
- P
- P → Q
- ---
- Q

If we know P is true

And we know P implies Q (if P then Q)

Then we can conclude Q is true

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Example For Modus Ponens:

- Given: (Joe is a father)
- And: (Joe is a father) → (Joe has a child)
- Conclude: (Joe has a child)

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Chain Rule

- Form $P \rightarrow Q$ and $Q \rightarrow R$, infer $P \rightarrow R$.
- Or
- $$\frac{P \rightarrow Q \qquad Q \rightarrow R}{P \rightarrow R}$$

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Example for Chain Rule

- Given: (programmer likes LISP)→ (programmer hates COBOL)
- **and** : (programmer hates COBOL) → (programmer likes recursion)
- **Conclude**: (programmer likes LISP)→ (programmer likes recursion)

- LISP → **Lis**t **P**rocessing
- COBOL → **Co**mmon **B**usiness **O**riented **L**anguage
- Prolog → **Programming in Log**ic

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Assignment-3

1. Construct a **truth Table** for the expression **(A & (A V B))**.
2. Determine whether each of the following sentences is
   (a) Satisfiable
   (b) Contradictory
   (c) Valid

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Assignment-3 Conti…

- (i) S_1: (P & Q) V ˜(P & Q)
- (ii) S_2: (P V Q)→ (P & Q)
- (iii) S_3: (P V Q)→ R V ˜Q
- (iv) S_4: (P V Q)& (PV ˜Q) V P
- (v) S_5: P → Q→ ˜P
- (vi) S_6: P V Q & ˜P V ˜Q &P

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- **Thanks**

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